ISLR 4.8.3

$$p_k(x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^{K} \pi_l f_l(x)}, \quad f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} e^{-\left(\frac{(x-\mu_k)^2}{2\sigma_k^2}\right)}$$

$$\log(\pi_k f_k(x)) = \log \pi_k - \log\sqrt{2\pi} - \log \sigma_k - \frac{(x-\mu_k)^2}{2\sigma_k^2}$$

($\log\sqrt{2\pi}$: same for all models)

$$\delta_k(x) = \log \pi_k - \log \sigma_k - \frac{(x-\mu_k)^2}{2\sigma_k^2}$$

$$\delta_k(x) = \log \pi_k - \log \sigma_k - \frac{(x^2 - 2x\mu_k + \mu_k^2)}{2\sigma_k^2}$$

$$\delta_k(x) = \log \pi_k - \log \sigma_k - \frac{x^2}{2\sigma_k^2} + \frac{\mu_k x}{\sigma_k^2} - \frac{\mu_k^2}{2\sigma_k^2}$$

quadratic with $a_k = -\frac{1}{2\sigma_k^2}$ $b_k = \frac{\mu_k}{\sigma_k^2}$, $c_k = -\frac{\mu_k^2}{2\sigma_k^2} + \log \pi_k - \log \sigma_k$

Decision boundary where $\delta_k(x) = \delta_j(x)$

$$a_k x^2 + b_k x + c_k = a_j x^2 + b_j x + c_j$$

$$(a_k - a_j)x^2 + (b_k - b_j)x + (c_k - c_j) = 0$$

$a_k - a_j = \left(-\frac{1}{2\sigma_k^2} + \frac{1}{2\sigma_j^2}\right) 2\sigma_k^2\sigma_j^2 = -\sigma_j^2 + \sigma_k^2 \neq 0$ Therefore quadratic decision boundary